

Building Java Programs

Scanner



Interactive programs

- We have written programs that print console output, but it is also possible to read *input* from the console.
 - The user types input into the console. We capture the input and use it in our program.
 - Such a program is called an *interactive program*.
- Interactive programs can be challenging.
 - Computers and users think in very different ways.
 - Users misbehave.

Input and `System.in`

- `System.out`
 - An object with methods named `println` and `print`
- `System.in`
 - not intended to be used directly
 - We use a second object, from a class `Scanner`, to help us.
- Constructing a `Scanner` object to read console input:
`Scanner name = new Scanner(System.in);`
 - Example:
`Scanner console = new Scanner(System.in);`

Java class libraries, import

- **Java class libraries:** Classes included with Java's JDK.
 - organized into groups named *packages*
 - To use a package, put an *import declaration* in your program.

- **Syntax:**

// put this at the very top of your program

```
import packageName.*;
```

- Scanner is in a package named `java.util`

```
import java.util.*;
```

- To use `Scanner`, you must place the above line at the top of your program (before the `public class` header).

Scanner methods

Method	Description
<code>nextInt()</code>	reads a token of user input as an <code>int</code>
<code>nextDouble()</code>	reads a token of user input as a <code>double</code>
<code>next()</code>	reads a token of user input as a <code>String</code>
<code>nextLine()</code>	reads a <i>line</i> of user input as a <code>String</code>

- Each method waits until the user presses Enter.
 - The value typed is returned.

```
System.out.print("How old are you? ");    // prompt
int age = console.nextInt();
System.out.println("You'll be 40 in " +
    (40 - age) + " years.");
```

- **prompt:** A message telling the user what input to type.

Example Scanner usage

```
import java.util.*;    // so that I can use Scanner

public class ReadSomeInput {
    public static void main(String[] args) {
        Scanner console = new Scanner(System.in);

        System.out.print("How old are you? ");
        int age = console.nextInt();

        System.out.println(age + "... That's quite old!");
    }
}
```

- Output (user input underlined):

```
How old are you? 14
14... That's quite old!
```

Another Scanner example

```
import java.util.*;    // so that I can use Scanner

public class ScannerSum {
    public static void main(String[] args) {
        Scanner console = new Scanner(System.in);

        System.out.print("Please type three numbers: ");
        int num1 = console.nextInt();
        int num2 = console.nextInt();
        int num3 = console.nextInt();

        int sum = num1 + num2 + num3;
        System.out.println("The sum is " + sum);
    }
}
```

- Output (user input underlined):

Please type three numbers: 8 6 13
The sum is 27

- The Scanner can read multiple values from one line.

Input tokens

- **token:** A unit of user input, as read by the `Scanner`.
 - Tokens are separated by *whitespace* (spaces, tabs, newlines).
 - How many tokens appear on the following line of input?

```
23   John Smith   42.0 "Hello world"   $2.50   " 19"
```

- When a token is not the type you ask for, it crashes.

```
System.out.print("What is your age? ");  
int age = console.nextInt();
```

Output:

```
What is your age? Timmy  
java.util.InputMismatchException  
    at java.util.Scanner.next(Unknown Source)  
    at java.util.Scanner.nextInt(Unknown Source)  
    ...
```

Scanners as parameters

- If many methods read input, declare a `Scanner` in `main` and pass it to the others as a parameter.

```
public static void main(String[] args) {  
    Scanner console = new Scanner(System.in);  
    int sum = readSum3(console);  
    System.out.println("The sum is " + sum);  
}
```

// Prompts for 3 numbers and returns their sum.

```
public static int readSum3(Scanner console) {  
    System.out.print("Type 3 numbers: ");  
    int num1 = console.nextInt();  
    int num2 = console.nextInt();  
    int num3 = console.nextInt();  
    return num1 + num2 + num3;  
}
```